When Does a Regulatory Graph Become a Mechanism? Prospective Perturbation Testing with TopoMIEG-FM

Abstract

Gene regulatory networks describe who may regulate whom, but a biological mechanism should also predict what happens after intervention. We ask a direct question: can a candidate regulatory graph predict the transcriptome change caused by knocking down one regulator? TopoMIEG-FM turns a signed transcription-factor–target graph and an unperturbed expression state into such a prediction. It anchors the graph to the baseline state, perturbs one regulator in silico, solves for the new state, and compares the predicted gene-expression change with an experiment that was not used to build the prediction. In 128 synthetic systems with known generating mechanisms, the correct mechanism was preferred to both higher-confidence incorrect alternatives and structurally matched hard decoys in all 128 cases. We then generated predictions for 39 K562 transcription-factor perturbations before examining genome-scale CRISPRi Perturb-seq outcomes. All 39 models converged numerically, but their predicted downstream changes showed no aggregate agreement with held-out responses (median Spearman $\rho = -0.045$; 95% bootstrap interval −0.171 to 0.080). Reconstructing the outcomes after count normalization gave the same conclusion. The targeted regulator decreased in all 29 cases in which it was measurable, whereas downstream responses showed little agreement across independent guides. We call the requirement that a graph predict an independent perturbation its mechanistic closure. The results show why this requirement matters: stable equations and plausible edges are not sufficient to establish a predictive biological mechanism.

Keywords: gene regulatory networks; perturbation biology; mechanistic inference; Perturb-seq; network dynamics; reliability

1. Introduction

A gene regulatory network describes which regulators may influence which genes. Such networks are reconstructed from expression, chromatin, sequence, and multi-omic data and are central to modern regulatory genomics [1–3]. A mechanism makes a stronger claim: it should also tell us how the system changes when we intervene. Perturbation experiments provide exactly this kind of evidence by measuring the consequences of deliberately altering a regulator [4–7].

These are three different questions. First, is a transcription-factor–target relation supported? Second, can many such relations coexist in a mathematically stable model? Third, does that model predict the expression change caused by perturbing one of its regulators? Evidence for the first two questions does not answer the third.

Existing methods address neighboring parts of this problem. Single-cell and multi-omic approaches infer increasingly detailed regulatory graphs [2,3]; Perturb-seq measures transcriptome-wide responses to genetic intervention [5–7]; CellOracle uses inferred networks for in silico transcription-factor perturbation [12]; and newer methods such as PSGRN use perturbational data to improve network inference [13]. Our focus is different: given a candidate graph that is already specified, how can we test whether its joint behavior is mechanistically credible?

Consider a transcription factor linked to twenty target genes. A stable network solution shows that the proposed interactions can coexist mathematically. A mechanistic test asks something more concrete: after the transcription factor is knocked down, do those twenty genes move in the directions and relative ordering predicted by the model?

We call this additional requirement mechanistic closure: a candidate graph must generate a perturbation-induced state change that agrees with an independent experiment. Baseline reconstruction alone is not evidence for closure, because the baseline is used to anchor the model. The test begins after intervention.

TopoMIEG-FM implements this test in four steps: take a signed candidate graph, anchor it to an unperturbed expression state, perturb one regulator in silico, and predict the resulting downstream expression change. We first evaluate the test in synthetic systems with known generating mechanisms. We then apply the same logic prospectively in K562 cells using DoRothEA TF–target relations [8], an independent ENCODE baseline [10], and held-out genome-scale CRISPRi Perturb-seq responses from Replogle et al. [7] distributed through scPerturb [9].

The two settings separate capability from real-data validity. In synthetic systems, the test distinguishes the generating mechanism from carefully matched alternatives. In K562 cells, all candidate models are numerically stable, yet their predicted downstream changes show no aggregate agreement with held-out perturbational responses, even after count normalization. The central result is therefore a separation between mathematical stability and experimentally supported mechanism.

2. From a regulatory graph to a testable perturbation prediction

TopoMIEG-FM evaluates a graph that is supplied in advance; it does not learn that graph from the perturbation outcomes. We represent the graph by a signed weight matrix W, where W_ij is the modeled effect of regulator j on target i. The unperturbed expression profile is mapped to a bounded state x_obs and used to set the model's baseline operating point:

$$b = \operatorname{atanh}(x_obs) - W\, x_obs, \qquad x_obs = \tanh(b + W\, x_obs).$$

By construction, the observed baseline is a fixed point. Baseline agreement therefore shows that the model has been anchored correctly; it is not the validation result. We then perturb source regulator s by an amount δ and solve for the new equilibrium:

$$b_pert = b - \delta\, e_s, \qquad x^*_pert = \tanh(b_pert + W\, x^*_pert).$$

The difference between the perturbed and baseline states is the predicted downstream expression change:

$$\Delta x_pred = x^*_pert - x_obs.$$

We compare Δx_pred with an independently measured response Δx_obs. The logic is deliberately sequential: edge evidence defines a candidate graph; fixed-point convergence shows that the graph has a mathematically coherent state; agreement with the intervention tests whether that state model predicts biological behavior. Dynamic network modeling has long emphasized behavior rather than connectivity alone [11]. Here, that principle becomes an explicit prospective test.

TopoMIEG-FM is therefore an evaluator of a candidate graph, not another graph-inference algorithm. Its output is a predicted perturbation-induced state change. In the synthetic experiments, where a target and expected direction are prespecified, we summarize that change by a signed response score R_eq = s_claim(Δx_target)/|δ|; larger values mean that the candidate produces a stronger response in the expected direction.

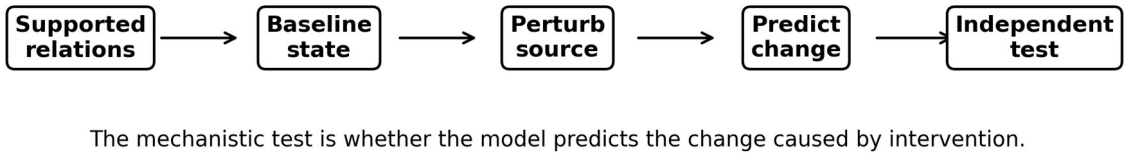


Figure 1 | From a regulatory graph to a perturbation test. A candidate graph is anchored to the unperturbed expression state, one regulator is perturbed, and the model predicts the resulting expression change. That prediction is then compared with an independent experiment.

3. Controlled systems show that the test can recover known mechanism-level structure

We first asked whether the test can recover known mechanism-level structure under controlled conditions. In synthetic systems, the generating graph is known, so we can compare its signed response score with alternatives designed to preserve increasingly strong forms of plausibility.

Across 128 synthetic worlds, the generating mechanism had a larger signed response score than a deliberately higher-confidence but incorrect alternative in every world. The median paired score difference was 0.0983 (95% bootstrap interval 0.0943–0.1025), and fixed-point residuals remained below 2.8×10^{-13}.

We next constructed a harder decoy that preserved the direct-edge magnitude, local degree, and total two-step path burden while disrupting signed path coherence. The generating mechanism again scored higher in 128/128 worlds, with a median paired difference of 0.0165 (95% interval 0.0162–0.0169). Context swaps and sign reversals were distinguished in 128/128 worlds, and the generating topology was preferred to a permuted alternative in 125/128 worlds.

The matched comparisons show that the synthetic test is sensitive to joint signed structure rather than simply edge confidence or graph size. They provide a controlled reference for the real-data experiment that follows.

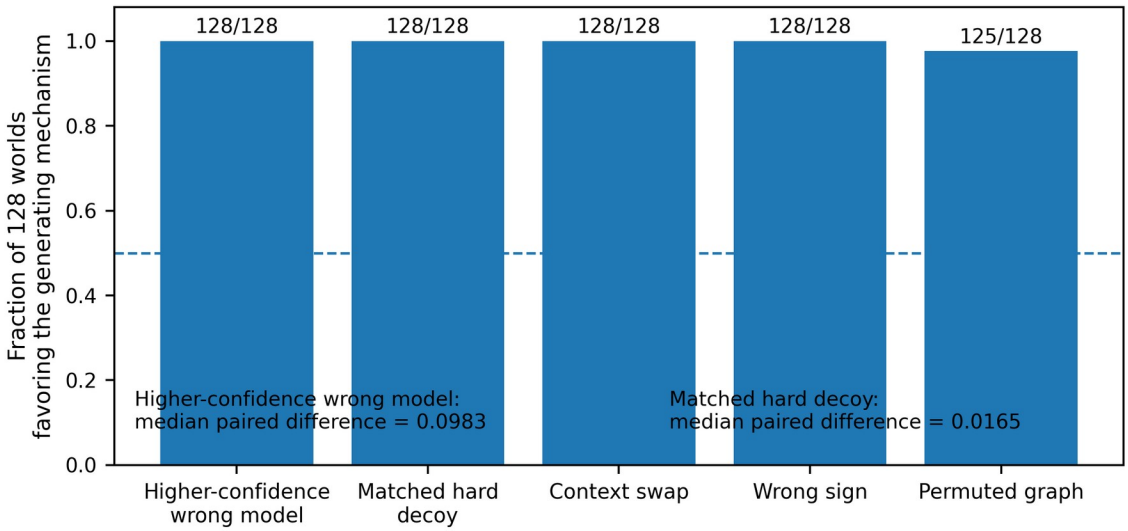


Figure 2 | Controlled systems establish identifiability. The generating mechanism is compared with alternatives that preserve different amounts of local evidence. Numbers above bars show the number of 128 synthetic worlds in which the generating mechanism is preferred.

4. Prospective K562 testing: stable models do not predict the aggregate response

4.1 Predictions were fixed before outcome access

For the real-data experiment, we built 39 candidate K562 regulatory models from DoRothEA confidence A/B signed TF–target relations [8] and an independent ENCODE K562 expression baseline [10]. Sources were selected without consulting the held-out Perturb-seq target list: each had to be expressed at baseline and have at least 20 baseline-expressed high-confidence targets. The 39 models produced 3,383 source–target predictions.

The source set, graph construction, perturbation strength, prediction vectors, and 100 topology rewires per source were fixed before perturbational expression outcomes were accessed. This design makes the K562 experiment a prospective test of a pre-existing prediction rather than a fit to the observed response.

Held-out responses came from the genome-scale K562 CRISPRi Perturb-seq study of Replogle et al. [7], using the harmonized object distributed by scPerturb [9]. For each transcription factor, we summarized the observed downstream expression change relative to non-targeting controls; the exact batch- and guide-level aggregation is described in Methods.

4.2 All models converged, but the aggregate correlation was near zero

All 39 candidate models converged to stable fixed points. The largest fixed-point residual was 2.39×10^{-12} and the largest effective spectral radius was 0.073, confirming that the numerical solutions themselves were stable.

The held-out predictive result was different. Across all 39 sources, the median Spearman correlation between predicted and observed downstream changes was −0.0455; a 2,000-replicate source bootstrap gave a 95% interval of −0.1713 to 0.0803. Source-level correlations varied in both directions, with no aggregate positive association.

We also compared the equilibrium prediction with direct one-step propagation, $\Delta x_{one} = -0.50\, W e_s$. One-step rank correlation was defined for 27 sources; in the other 12, the one-step vector was constant across scored genes. Among the 27 evaluable sources, the median equilibrium-minus-one-step difference was −0.0015 (95% interval −0.1584 to 0.0675), providing no aggregate advantage for iterative equilibrium propagation.

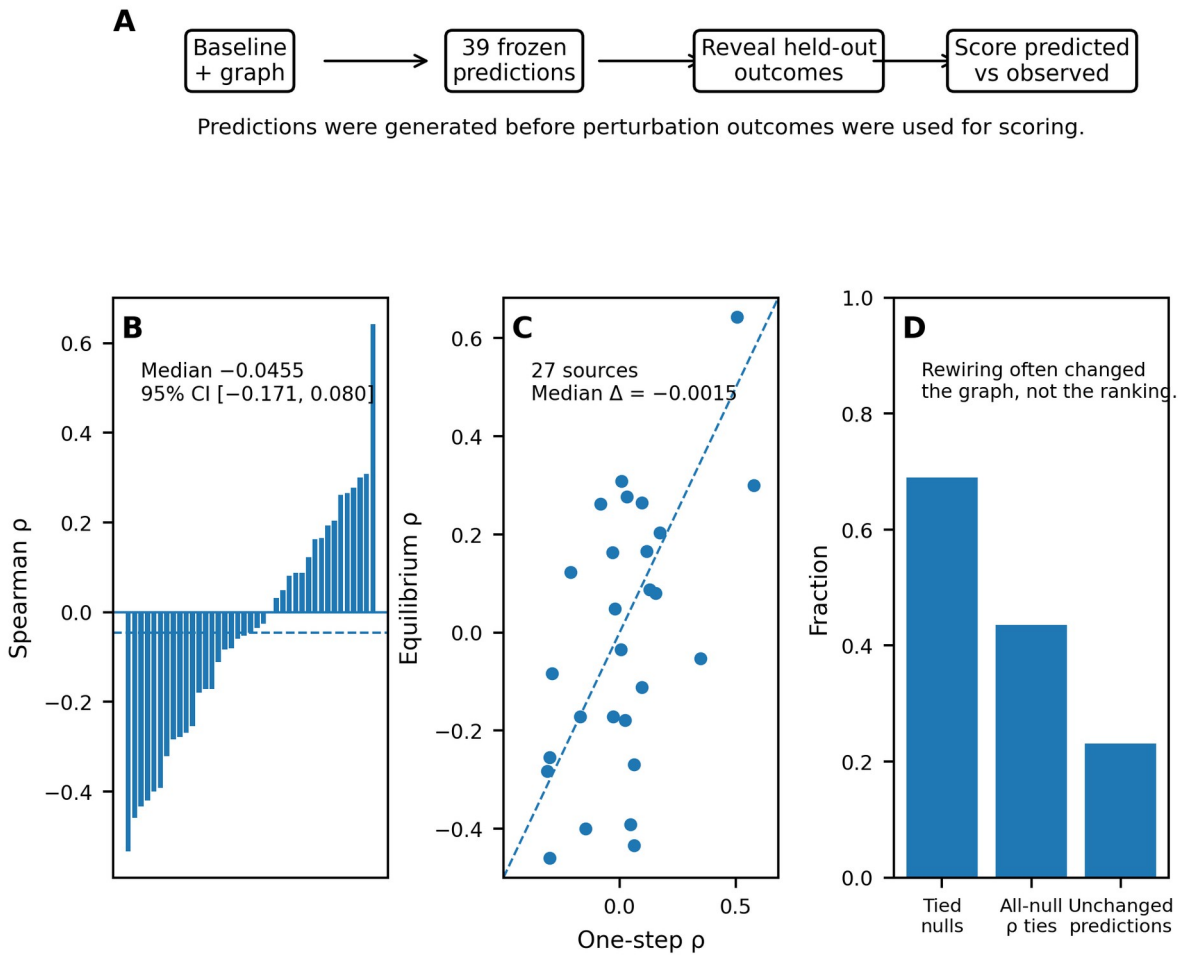


Figure 3 | Prospective K562 perturbation test. (A) The model and 39 state-transition predictions were fixed before held-out CRISPRi outcomes were used for scoring. (B) Source-level Spearman correlations between equilibrium predictions and observed downstream changes. (C) Equilibrium versus one-step correlations for the 27 sources with nonconstant one-step predictions. (D) Degree- and sign-preserving graph rewiring often had little effect on the predicted ranking, limiting the discriminability of this structural control.

The rewiring control was only weakly discriminating because changing the graph often left the prediction nearly unchanged. Across sources, the median fraction of rewired correlations tied with the native correlation was 0.69; 17 of 39 sources produced the same correlation across all 100 rewires, and nine produced unchanged prediction vectors across all rewires. Structural controls therefore need to be checked in prediction space, not only in graph space.

5. Target knockdown is consistent, but downstream responses are weakly reproducible

We next separated three components of the K562 experiment: whether CRISPRi reduced the intended regulator, whether independent guides produced similar downstream expression changes, and whether the result depended on how the count matrix was normalized.

5.1 The targeted regulators moved in the expected direction

Twenty-nine of the 39 perturbed transcription factors were themselves among the 8,248 measured genes, and all 29 showed lower expression after CRISPRi (Fig. 4A). Direct source knockdown was therefore visible wherever the source gene was measured.

5.2 Independent guides produced weakly concordant downstream states

Eleven sources had multiple guide constructs. Their guide-specific downstream response vectors agreed poorly: the median pairwise Spearman correlation was 0.004, with some sources positive and others near zero or negative (Fig. 4B). Thus, reproducible source knockdown did not imply a reproducible transcriptome-wide response.

5.3 Count normalization did not change the overall conclusion

Because the distributed H5AD matrix contained UMI counts, we repeated the analysis after standard per-cell normalization to 10,000 counts followed by log1p transformation. All source definitions, prediction vectors, and aggregation rules were kept unchanged.

Normalization did not change the overall pattern. Across 39 sources, the median equilibrium Spearman correlation was −0.0603 (95% interval −0.1169 to 0.0490). The median equilibrium-minus-one-step difference among the 27 evaluable sources was −0.0554 (95% interval −0.1240 to 0.0825). Raw and normalized observed-response vectors remained strongly related (median source-level $\rho = 0.853$), while multi-guide reproducibility remained near zero (median $\rho = 0.011$) and all 29 measurable source genes still decreased.

The pattern is therefore simple: CRISPRi consistently reduced the targeted regulator where it could be measured, but downstream transcriptome changes were weakly reproducible across guides, and the equilibrium model did not predict their aggregate direction under either expression representation.

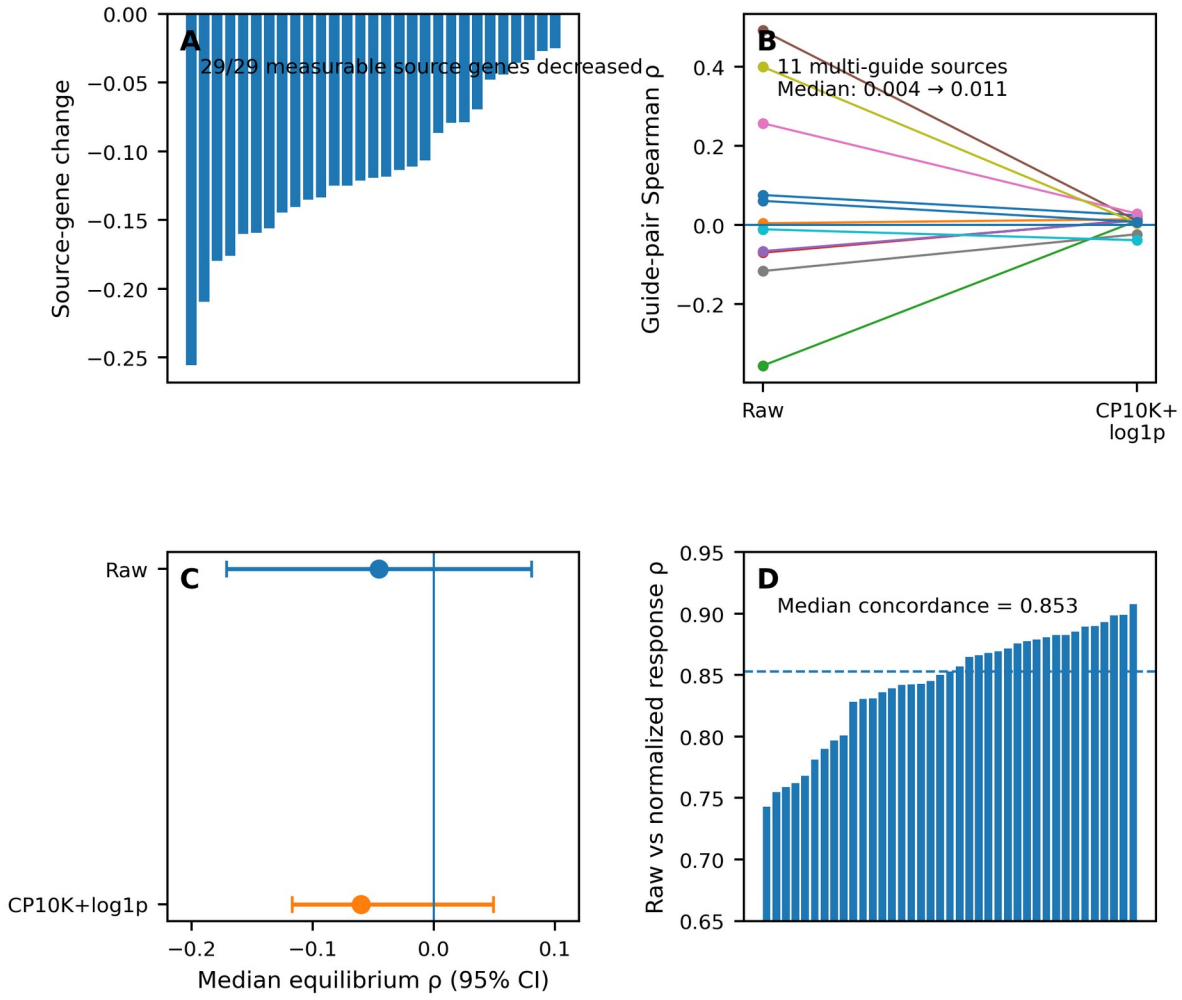


Figure 4 | Distinct empirical layers of the K562 experiment. (A) All 29 measurable source genes decrease after CRISPRi. (B) Downstream state-transition vectors show little agreement across independent guides; normalization does not materially improve concordance. (C) The aggregate equilibrium correlation remains centered near zero under raw-count and CP10K-log1p outcome construction. (D) Raw-count and normalized response vectors remain substantially correlated across the 39 sources.

Together, the K562 data separate three quantities that should not be conflated: whether the intended regulator is perturbed, whether the downstream response is reproducible, and whether a regulatory model predicts that response. The first is strong here; the latter two are weak.

6. Discussion

This study separates mathematical stability from experimentally supported mechanism. In synthetic systems, known mechanism-level structure produces a recognizable perturbational signature and the framework recovers it. In K562 cells, the same equations converge cleanly, but the predicted downstream changes do not reproduce the held-out response.

This distinction matters because modern regulatory genomics now has both detailed candidate networks [1–3] and large-scale perturbation readouts [5–7]. These evidence types answer different questions. Network inference proposes regulatory structure; perturbation measures system behavior. A mechanistic claim becomes stronger when the proposed structure makes a prediction before the intervention outcome is examined.

The synthetic results show that this test can succeed when informative joint structure is present. It distinguishes the generating model from alternatives with higher edge confidence or closely matched local structure, and it is sensitive to context and sign. The relevant signal therefore lies in the behavior of the assembled system, not simply in whether individual edges look plausible.

The K562 experiment shows why the perturbation readout itself also needs to be characterized. Direct regulator knockdown was consistent, whereas transcriptome-wide responses varied across guide constructs. Perturbation efficacy, downstream reproducibility, and model agreement are therefore distinct quantities. This complements scPerturb's emphasis on quantifying perturbation effects and harmonizing perturbational datasets [9].

The rewiring analysis gives a parallel lesson for computational controls. Randomizing a graph is informative only when the randomization changes the prediction being evaluated. In this experiment, many degree- and sign-preserving rewires left the equilibrium ranking nearly unchanged. Structural controls should therefore be evaluated for separation in prediction space as well as graph space.

TopoMIEG-FM is therefore a test of mechanistic claims rather than a network-discovery algorithm. It asks three increasingly strong questions: Are the relations supported? Can they form a stable joint model? Does that model predict an independent intervention? A stable solution answers the second question. Agreement with perturbation answers the third. In practical terms, mathematical closure generates a mechanistic hypothesis; prospective perturbational agreement provides the experimental test.

7. Methods

7.1 Regulatory relations and K562 baseline

Human signed TF–target relations were obtained from DoRothEA version 1.23.0, using confidence levels A and B only [8]. For each source transcription factor, the candidate node set contained the source and its baseline-expressed direct targets. All A/B DoRothEA interactions among those nodes were included, allowing target-to-target interactions and feedback where supported. Edge signs followed the reported mode of regulation and all included edges had unit magnitude before global scaling.

The baseline K562 expression state was constructed from ENCODE total RNA-seq experiment ENCSR885DVH [10], using replicates ENCFF424CXV and ENCFF073NHK. Gene-level values were mapped to HGNC symbols, producing 15,360 unique baseline genes. For the dynamical state, expression ranks were mapped to $(-1,1)$ using $x_i = 2(\mathrm{rank}_i - 0.5)/n - 1$ and clipped to $[-0.95, 0.95]$.

7.2 State model and perturbation prediction

For each source-specific model, the signed adjacency operator W was scaled to spectral norm 0.30. The baseline intercept was $b = \operatorname{atanh}(x_{obs}) - W x_{obs}$, which makes x_{obs} a fixed point of $x = \tanh(b + W x)$. A source perturbation was represented by $b_{pert} = b - 0.50\, e_s$. The perturbed fixed point was solved iteratively and the predicted response was $\Delta x_{eq} = x^*_{pert} - x_{obs}$. The source gene itself was excluded from downstream scoring.

The direct comparator was the one-step response $\Delta x_{one} = -0.50\, W e_s$. Numerical admissibility required fixed-point residual $\leq 10^{-10}$ and effective spectral radius ≤ 0.35. The observed maximum values were 2.39×10^{-12} and 0.073, respectively.

7.3 Prospective source set and prediction generation

The source universe was determined without using the Perturb-seq target list. A transcription factor was eligible if it was expressed in the independent baseline and had at least 20 baseline-expressed A/B targets. Thirty-nine sources met this rule. Predictions were generated for all 39 sources before held-out expression outcomes were materialized. One hundred degree- and sign-count-preserving rewires were generated per source using fixed seed 20261107. The source set, graph construction, perturbation magnitude, prediction vectors, and rewired predictions were fixed before outcome scoring.

7.4 Held-out Perturb-seq outcomes

Held-out single-cell responses were taken from the K562 genome-scale CRISPRi Perturb-seq experiment of Replogle et al. [7] through the harmonized scPerturb H5AD distribution [9]. The matrix contained 1,989,578 cells and 8,248 measured genes. Explicit controls were cells annotated gene = non-targeting, perturbation = control, and nperts = 0 (75,328 cells). Perturbed cells had gene equal to the frozen source and nperts = 1.

7.5 Outcome aggregation and scoring

For source s, guide g, and batch b, the observed batch response was the mean expression of perturbed cells minus the pooled explicit-control mean from the same batch. Batch responses were averaged with equal weight within guide, and guide responses were averaged with equal weight within source. Cells therefore did not determine cross-batch or cross-guide weights. Gene matching used exact symbols and the source gene was excluded from primary downstream scoring.

The primary source-level statistic was Spearman correlation between Δx_eq and the observed downstream response. Aggregate performance was summarized by the median across source perturbations. Uncertainty was estimated with 2,000 source-level bootstrap replicates using seed 20261108. The equilibrium-versus-one-step comparison used the paired difference in source-level Spearman correlations where both were defined.

For the topology control, each native source correlation was compared with the distribution of 100 rewired predictions. We additionally quantified whether rewiring materially changed the prediction vector, because a structural randomization that leaves the scored output unchanged cannot provide a discriminating null.

7.6 Expression-representation sensitivity

The distributed H5AD X matrix was float32 but integer-valued and its sampled row sums matched obs/ncounts exactly, indicating UMI counts. We therefore repeated outcome construction after per-cell scaling to 10,000 total counts followed by log1p transformation. Control definitions, batch matching, guide averaging, source averaging, gene matching, predictions, and scoring rules were unchanged.

7.7 Controlled synthetic experiments

Synthetic evaluations used 128 independent worlds per comparison. The first comparison tested the generating mechanism against a deliberately higher-confidence incorrect alternative. A harder direct-edge decoy preserved signed direct-edge magnitude, motif degree, and absolute two-step burden while disrupting local signed-path coherence. Additional tests swapped context, reversed signs, or permuted topology. Fixed-point convergence and numerical stability were checked for every world.

7.8 Data and code availability

The K562 perturbation data are from Replogle et al. [7] and the harmonized scPerturb resource [9]. Baseline K562 RNA-seq is available from ENCODE [10], and TF–target relations are from DoRothEA [8]. Analysis code, frozen prediction manifests, configuration files, and reproducibility instructions accompany the submission repository.

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